Is There Exploitable Concept Drift in Industrial Tabular Data? A Pre-Registered Identifiability Audit

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Paper under double-blind review

Abstract

Which kind of temporal shift a tabular dataset exhibits—a changing rule $P(y \mid x)$ or moving covariates $P(x)$—is rarely measured, and we show the verdict depends on the measuring instrument. We study *staleness harm*, a deployment-native probe: does adding old examples to a fixed recent training set hurt future performance? Three executed failure modes make the naive probe untrustworthy. Label-noise decay under a provably fixed rule mints a "concept drift" verdict statistically matching a real industrial positive (+0.021 vs +0.024); the separability gate that should certify nulls saturates under duplicate rows and entity cohorts; and the concept/covariate separation is hypothesis-class-relative—kNN and linear probes false-fire where tree ensembles read correctly. We repair the instrument—a cross-fitted *denoised staleness* arm with a noise gate and a mapped abstention envelope, group-aware separability, and learnability-gated injection certificates—and validate it on a 14-cell pre-registered battery that includes rule change and noise drift co-occurring. A pre-registered audit of eight industrial datasets (TabReD) with confirmatory fresh-seed replication (10/10 stable) then yields an identifiability *map*, not a detection table: **no industrial dataset shows exploitable mean-rule drift above its detectable floor (0/8)**; the sole robust positive is designed drift; the one prior industrial positive is *diagnosed* by the instrument itself as label-noise decay; every remaining null carries a certificate—verified, blind, or refused. Anchor streams fix the sensitivity profile: monotone and single-switch rule changes fire 9/9, and recurring regimes are correctly silent with a negative-recency fingerprint. We release the instrument, battery, and audit trail, and argue that drift-type attribution without identifiability certificates—the current default in drift monitoring—is unreliable.

1 Introduction

Time-based splits on industrial tabular data collapse elaborate time-aware deep methods while gradient-boosted trees survive (TabReD; Rubachev et al., 2025). The standard motivation for time-aware architecture is *concept drift*: if the labeling rule $P(y \mid x)$ changes over time, a model that indexes, retrieves, or modulates by time should beat one that merely receives time as a feature. But whether the rule actually changes on these benchmarks—as opposed to the covariates $P(x)$ moving under a fixed rule—has not been measured, and our central claim is that measuring it is harder than the field assumes: **the concept/covariate verdict is a property of the measuring instrument (its hypothesis class, its noise assumptions, its window geometry) as much as of the data.** We make the measurement itself the object of study, in the setting a deployed system actually faces: train on the past, predict the future, decide what to do with old data.

The probe. We start from a deployment-native quantity we call *staleness harm*:

$$\text{staleness} = \text{score}(\text{train on recent } N) - \text{score}(\text{train on recent } N \cup \text{old } N), \quad \text{evaluated on a future window,}$$

computed rolling-origin over K timestamp-value windows. Because the recent portion is identical in both arms, covariate *coverage* cancels by construction: the only difference is N extra old examples. If the rule is fixed, old examples carry correct labels and adding them should not hurt; if the rule has changed, old labels contradict the current rule and pollute the fit—staleness > 0. This construction (unlike train-window selection, which uses the same mechanics for *adaptation*; Klinkenberg & Joachims, 2000) attributes *why* old data helps or hurts, and it needs no importance weights, no density ratios, and no overlap between window supports to compute—which is exactly what makes it attractive for industrial feature spaces where overlap-based decompositions degenerate (D'Amour et al., 2021; Cai et al., 2023).

Three executed failure modes. The naive form of this argument—"correct labels can only help"—is false for finite-capacity empirical risk minimization, and we show it fails silently in three distinct ways, each demonstrated by a constructed null that the probe misreads:

1. **Label-noise drift (the diagnosis that motivated this paper).** A regression stream with a *fixed* conditional mean and label-noise standard deviation shrinking $1.5 \to 0.3$ over time mints a "concept" verdict at staleness $+0.021$ $[+0.020, +0.023]$—statistically indistinguishable from a real industrial positive we had previously obtained ($+0.024$ $[+0.018, +0.030]$ on sberbank-housing). Old labels need not be *wrong* to hurt; being *noisier* suffices, because unweighted ERM is not efficient under heteroscedasticity. A second channel—x-dependent noise whose scale decays—fires the raw probe at $+0.025$ under an equally fixed rule.

2. **Overlap-gate saturation.** The natural certificate for "a null here is uninformative" is a window-separability AUC (can a classifier tell old rows from future rows?). Computed with a row-level train/test split, this gate saturates to 0.994–1.000 under exact duplicates, tight near-duplicates, and entity cohorts with *zero* covariate shift—the classifier memorizes rows, not distributions. Eight of ten real datasets read exactly 1.000 under the naive gate.

3. **Hypothesis-class relativity.** Rerunning the probe's own ground-truth suite with the model class swapped shows the concept/covariate *separation* is not a property of the probe design: a kNN probe reads pure covariate shift as concept drift ($+0.098$) and a linear probe reads prior shift as concept drift ($+0.026$), while HGB and random forests pass all controls. This is the empirical face of known theory (Shimodaira, 2000; Hinder et al., 2023): under misspecification, the ERM optimum depends on the training covariate distribution, so "old data hurts" can be manufactured by geometry alone.

The repaired instrument. Each failure mode gets a repair that is itself validated by execution (§3–4): a *denoised staleness* arm that replaces old labels with cross-fitted within-old-window model predictions—under noise-only drift the pseudo-labels are approximately correct and the harm vanishes; under a changed rule they still encode the old rule and the harm persists—plus a per-window *noise gate* with an explicitly mapped validity envelope beyond which the instrument abstains; a group-aware, size-matched separability estimate that deflates the memorization channel while leaving honest drift intact; and *learnability-gated injection controls* that turn "this dataset is unmeasurable" from an assumption into a demonstration: a known rule rotation is planted in the dataset's own covariate geometry, verified to be learnable in-window, and only then is a persistent null allowed to certify blindness. All verdicts are explicitly scoped to the tree-ensemble hypothesis class, with linear and kNN probes carried as *canaries* whose false fires are reported, not hidden.

The audit. We pre-registered the decision cascade, thresholds, seed protocol, and aggregate reading before the runs (rule $\to$ prediction $\to$ execution $\to$ read, evidenced by commit timestamps), then audited eight TabReD datasets, Electricity, and the INSECTS streams, with a confirmatory fresh-seed replication and a model-class panel. The result is not a detection table but an identifiability map (§5): industrial mean-rule drift above the per-dataset detectable floor is **0/8**; the designed-drift stream is the sole positive (and its denoised staleness *exceeds* the raw one—the signature of genuine rule change); the single prior industrial positive is *diagnosed*—not merely retracted—as label-noise decay, by the same instrument that once minted it; two datasets earn a verified no-concept certificate via injection recovery; one earns a blindness certificate; three fail to earn one (their injections are unlearnable—a fact the naive protocol would have laundered into "earned blindness"). Anchor streams (§6) give the instrument a coherent sensitivity profile—monotone and single-switch rule changes fire 9/9; recurring regimes are correctly silent and are flagged by a negative recency gain, which is impossible under one-way drift; malware (EMBER) reads as coverage-driven decay, not label rot, consistent with how temporal degradation actually arises there (Pendlebury et al., 2019).

Contributions.

1. *An executed anatomy of drift-type attribution failure* (§3–4): three constructed nulls that silently mint concept-drift verdicts (noise decay, gate memorization, class relativity), each with magnitudes matching real borderline positives.

2. *A repaired, abstaining instrument* (§3): denoised staleness + noise gate with a mapped validity envelope; group-aware separability; learnability-gated injection certificates; class-scoped verdicts with canaries. Validated on a 14-cell pre-registered battery including the adversarial combinations (rule change *and* noise drift co-occurring must still fire—it does, at 58% of clean power).

3. *A pre-registered identifiability map of industrial tabular ML* (§5): 0/8 exploitable mean-rule drift; one diagnosed false positive with mechanism; per-dataset certificates and detectable-effect bounds; 10/10 confirmatory stability and 10/10 cross-class (HGB↔RF) agreement, with a real-data demonstration that a linear monitor flips the canonical Electricity dataset to "concept drift".

4. *A sensitivity profile with anchors* (§6): what this lens can and cannot see, established on designed-drift streams (fires iff old labels contradict the *current* regime), with the negative-recency fingerprint for recurring regimes and an honest malware null.

What we do not claim. We do not claim concept drift is absent from industrial tabular data—several cells are certified *blind*, and blindness is not absence. We do not claim the identifiability theory is new—that overlap failure blocks nonparametric identification is classical (D'Amour et al., 2021; Ben-David et al., 2010), and the class-relativity principle is established (Hinder et al., 2023); our contribution is the executed instrument, its failure anatomy, and the certified map. And we do not claim model-agnosticism: every verdict is relative to the deployed hypothesis class, which we argue is the only honest way to state drift-type attribution at all.

2 Related Work

Shift-type maps on tabular data. WhyShift (Liu et al., 2023) is the closest prior: it attributes performance drops to $Y \mid X$- vs X-shift across tabular datasets and concludes $Y \mid X$-shifts are prevalent—on predominantly spatial/subpopulation axes, under an overlap-assuming decomposition. Our delta is three-fold: the *temporal* axis on deployed industrial representations (where positivity actually fails and their lens degenerates), an *abstention arm* with per-dataset certificates (WhyShift assumes the decomposition is measurable), and the instrument-failure anatomy. DISDE (Cai et al., 2023) supplies the decomposition formalism we inherit terminology from; its shared-distribution term is identified only on common support—the boundary our map makes operational. TabReD itself (Rubachev et al., 2025) frames everything as "gradual temporal shift" and contains no decomposition or recency experiments; TableShift (Gardner et al., 2023) and Wild-Time (Yao et al., 2022) are shift benchmarks without drift-type attribution.

Identifiability and class-relativity. That $P(y \mid x)$ is not identified off-support without assumptions is textbook positivity (D'Amour et al., 2021, for high-dimensional overlap failure; Ben-David et al., 2010, for the learning-theoretic impossibility). Hinder et al. (2023) prove constructively that loss-based drift detection is class-relative in both directions—virtual drift that moves the loss, real drift invisible to it—and their survey (Hinder et al., 2024) states "the used model class is crucial." Loog et al. (2019) show ERM risk can be non-monotone in sample size even i.i.d., closing the door on any unconditional "more correct data cannot hurt" lemma. Gower-Winter et al. (2026) argue drift detection is ill-posed through windowing choices. We treat all of this as settled theory that the applied drift-monitoring stack has not absorbed, and contribute the executed demonstration + the repaired protocol; we state no theorems of our own—where the paper sounds formal (the envelope, the certificates), the content is a measured boundary, not a proved one.

Old data harming. Shimodaira (2000) is the mechanism for the misspecified case; the Data Addition Dilemma (Shen et al., 2024) documents mixture harm empirically in clinical ML; Klinkenberg & Joachims (2000) use held-out error over candidate training windows to *adapt* window size. We are not aware of prior work that uses the add-old-data contrast as a drift-*type* identifier with verdict semantics, or that reports the label-noise-decay false-positive channel—under the field-standard definition of real drift as any change in $P(y \mid x)$ (Gama et al., 2014; Webb et al., 2016), noise drift *is* drift, which is precisely why an instrument that claims to detect *exploitable rule change* must separate the two; we separate them by construction and validate the separation adversarially.

Pseudo-labels, denoising, cross-fitting. The denoised arm's mechanics are deliberately borrowed: replacing labels with model predictions is self-training (Scudder, 1965; Lee, 2013) and distillation (Hinton et al., 2015); the view of noisy labels as recoverable by a fitted model underlies noisy-label learning (Natarajan et al., 2013; Han et al., 2018); strictly out-of-fold prediction to avoid own-fit contamination is cross-fitting (Chernozhukov et al., 2018). What we take from these literatures is the estimator; the *use*—as the discriminating arm of a drift-type verdict, with an adversarially mapped validity envelope and an abstention rule—is the contribution we claim, and we scope it to its evidence: none of these adjacent literatures, to our reading, reports the failure channel this arm exists to defuse.

Drift detectors and their evaluation. Classical detectors (surveys: Gama et al., 2014; Lu et al., 2019) monitor loss or distribution statistics and are evaluated by injecting known drifts into streams (Poenaru-Olaru et al., 2022); Detectron (Ginsberg et al., 2023) defines harmful shift model-relatively. Our injection control differs in role: it is an *in-situ, per-dataset power certificate* (planted in the real covariate geometry, learnability-gated), used to distinguish earned blindness from instrument vacuity—a distinction we show matters on real data, where three of four "unidentifiable" cells fail the learnability gate.

Malware. TESSERACT (Pendlebury et al., 2019) established temporally honest evaluation in malware and documents performance decay. Our EMBER read (old data *helps*; staleness −0.012; detectable floor 0.0014) refines rather than contradicts it: the decay is coverage-driven (new families appear—covariate expansion), not label rot (a 2017 malware sample is still malware), which is exactly the distinction a deployment lens should draw.

3 The Instrument

Reading aid. One real cell exercises most of the vocabulary in Table 1. On sberbank-housing (§5.3), the raw arm fires (+0.024) but the noise gate is on (2.1×) and the denoised arm is negative—so the cascade returns NOISE-DRIFT-CONFOUNDED rather than concept; had all three aligned inside the envelope, the verdict would have been DEPLOYMENT-CONCEPT, subject to an injection positive control.

3.1 Setting and estimand

Rows (x_i, y_i, t_i) arrive over time; the auditor sees the full labeled history (retrospective audit, not online prediction). Fix K windows by timestamp-value rank (ties never straddle boundaries; per-seed boundary jitter). For each future window W_j (back half), with old anchor W_0 (earliest window with ≥ 200 rows and a trainable label set) and recent window W_{j-1}, draw size-matched samples ($N = \min(|\text{recent}|, |\text{old}|, 6000)$) and train three models of the deployed class: recent-only, old-only, recent $\cup$ old. Report per-seed means over future windows of

- **decay** = score(old model, held-out W_0) − score(old model, W_j)—aging, mechanism-blind;
- **recency gain** = score(recent) − score(old) on W_j—adaptation value, conflates coverage and rule change; *its sign is itself diagnostic (§6): negative recency is impossible under one-way drift and fingerprints recurring regimes*;
- **raw staleness** = score(recent) − score(recent $\cup$ old) on W_j—the attribution probe;

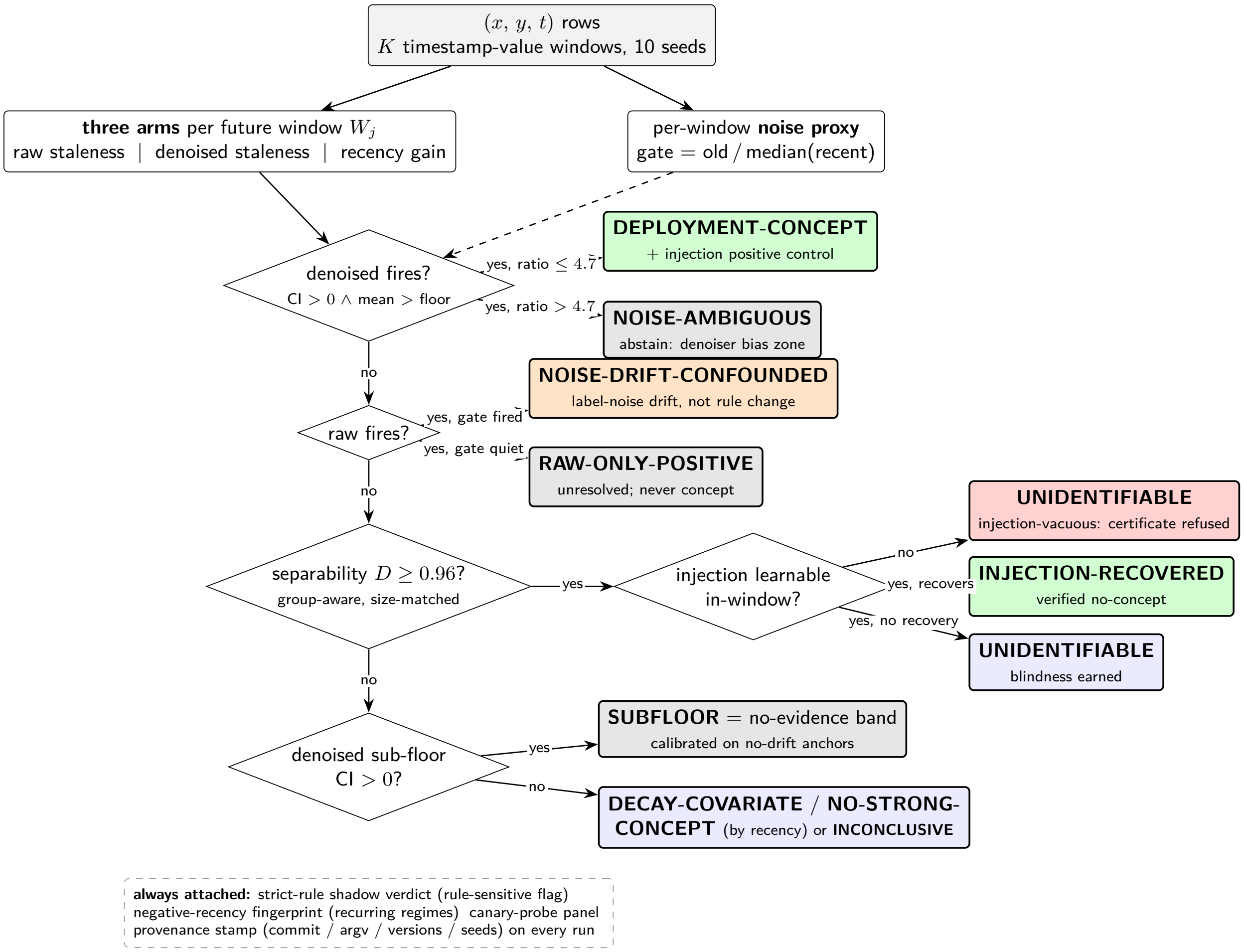

Figure 1: The decision cascade of the instrument. The cascade is specified verbatim in §3.3 and PREREG §3. Side channels attached to every run: the strict-rule shadow verdict (rule-sensitive flag), the negative-recency fingerprint for recurring regimes (§6), the canary-probe panel (§4.3), and provenance stamping on every run.

- **denoised staleness** = score(recent) − score(recent $\cup$ $(X_{\text{old}}, \hat{g}_{\text{old}}(X_{\text{old}}))$) on W_j, where $\hat{g}_{\text{old}}$ is a 2-fold cross-fitted model *within* W_0 producing strictly out-of-fold hard pseudo-labels.

Scores: AUC (binary), accuracy (multiclass), −RMSE on z-scored targets (regression). CIs are Student-t over 10 seeds; each seed re-draws a 90% row subsample, re-jitters window boundaries, and re-samples all training sets. These are seed-level intervals over heavily overlapping subsamples and are anti-conservative for tiny effects (§6 calibrates the resulting no-evidence band on no-drift anchors); verdict-level inference does not rest on them alone—the confirmatory fresh-seed replication (§5.1) is the operative stability check, and per-seed values are emitted with every run so that alternative constructions (window-block bootstrap, split-half) can be applied without re-execution. The pre-registered estimand is *exploitable mean-rule drift*: a change in the decision-relevant functional of $P(y \mid x)$ that makes old labels contradict the current rule for the deployed hypothesis class. This is deliberately narrower than the field-standard "any change in $P(y \mid x)$" (Webb et al., 2016)—the narrowing is the point, because the wider definition classifies label-noise drift as concept and thereby licenses retraining that cannot help.

Why the denoised arm separates rule change from noise drift. Under noise-only drift the conditional mean/Bayes rule is unchanged, so $\hat{g}_{\text{old}}$ estimates the *same* rule from noisy labels; its pseudo-labels are approximately correct denoised labels, and adding $(X_{\text{old}}, \hat{g}_{\text{old}}(X_{\text{old}}))$ to the recent set cannot systematically hurt—executed: the +0.021 noise-decay false positive collapses to +0.004 while the gate (below) flags

Table 1: Terminology (one line each).

term	meaning
raw staleness	future-window score(recent) − score(recent ∪ old); > 0 = old data hurts
denoised staleness	same, with old labels replaced by cross-fitted within-old-window predictions; > 0 = the *rule* changed
noise gate	old-window noise proxy / recent median; > 1.5 = label-noise drift present
envelope	noise-ratio 4.7, the measured boundary of denoiser validity; above it the instrument abstains
D	window-separability AUC (group-aware, size-matched); a routing statistic, *not* support overlap
injection	a known rule rotation planted in the dataset's own geometry; the per-dataset power certificate
earned blindness	injection learnable in-window yet unrecoverable → the geometry genuinely hides this signal class
injection-vacuous	injection unlearnable → the null certifies nothing
no-evidence band	sub-floor CI-positive readings; calibrated on no-drift anchors as noise
canary probe	a deliberately misspecified probe class (linear/kNN) whose false fires are reported
rule-sensitive	verdict differs between the primary and strict decision rules; barred from headlines

the mechanism. Under genuine rule change, $\hat{g}_{\text{old}}$ consistently estimates the *old* rule; its pseudo-labels still contradict the current rule, and the harm persists—executed: a rotating-rule stream keeps +0.541 of its +0.546 raw signal, and rule-change-plus-noise-decay still fires at +0.316. The denoiser is not unbiased: its pseudo-label error grows with old-window noise, and we *mapped the boundary*—at an old/recent noise ratio ≈ 5.7 the denoised arm itself crosses the decision floor on a pure null (+0.026). The instrument therefore carries an envelope constant (abstain above ratio 4.7) rather than a pretense of a theorem; per Loog et al. (2019), no unconditional soundness lemma exists to be had.

The noise gate. Per window, a fresh model of the deployed class is fit on 70% and its held-out irreducible-error proxy recorded (regression: MSE on z-scored y; binary: 1−AUC; multiclass: 1−accuracy). The gate statistic is old-window proxy / median recent-window proxy; it fires above 1.5 (stable synthetic controls calibrate at 0.75–0.99) and defines the envelope above 4.7. The gate alone is *not* a sufficient repair—a gate-veto rule would misfile genuine rule change co-occurring with noise drift (executed: +0.316 must fire despite gate 3.67)—it supplies the *mechanism label* (NOISE-DRIFT-CONFOUNDED) when the raw arm fires and the denoised arm does not.

3.2 Certificates: separability, injection, learnability

Window separability D (not "overlap"). D is the median held-out AUC of a classifier separating W_0 from each back-half window, on the proxy-stripped feature space, with two repairs over the naive version: size-matching by *random subsample* (a head-slice on time-sorted rows manufactures separability from window-size imbalance—executed: shuffle-D 0.94 with no static feature, collapsing to 0.50 under the fix) and a *group-aware* train/test split keyed on near-duplicate clusters (z-scored rows rounded to 0.1): exact duplicates at multiplicity 5 inflate row-split D to 0.994 and entity cohorts to 1.000 with zero covariate shift; the group split returns both to ≈ 0.50 while honest drift keeps D high. We report D as *separability*, never as support overlap: a single genuinely predictive drifting feature saturates it, so $D \geq D^* = 0.96$ routes a staleness null to the injection control rather than concluding anything by itself.

Learnability-gated injection. For any candidate-unidentifiable dataset (and, as a positive control, for any CONCEPT verdict), a reference concept—a rule rotation of fixed strength on the two highest-variance features—is planted in the dataset's own (X, t) geometry and the full staleness pipeline re-run (10 seeds, same power as the main read). Before the null is allowed to mean anything, the injected rule must be *learnable in-window* (held-out AUC ≥ 0.65 / $R^2 \geq 0.20$ / accuracy $\geq$ majority + 0.10): executed on a constructed geometry whose top-variance features are heavy-tailed junk, the injected rule is unlearnable (in-window AUC 0.506) and the naive protocol grants "earned blindness" vacuously; the gate converts that to *injection-vacuous*—certificate refused. Outcomes: recovery (⇒ the real null was informative: verified no-

Table 2: The 14-cell pre-registered battery. Verdicts under the full cascade; all cells $n = 12{,}000$, $d = 10$, $K = 10$, 5 seeds.

cell (truth)	raw	denoised	gate	verdict
rotating rule (binary)	+0.280	+0.289	0.90	**CONCEPT** ✓
rotating rule + drifting nuisance	+0.203	+0.206	1.18	**CONCEPT** ✓ (proxy-stripped)
rotating rule (regression)	+0.546	+0.541	0.91	**CONCEPT** ✓
rotating rule + noise decay	+0.357	**+0.316**	3.67 (fires)	**CONCEPT** ✓—the gate must not veto
covariate shift, fixed rule	−0.001	−0.000	0.74	UNIDENT (blindness earned) ✓
stationary	−0.001	−0.000	0.75	NO-STRONG-CONCEPT ✓
prior shift, fixed rule (multiclass)	−0.003	−0.004	0.84	UNIDENT, injection-vacuous ✓
mild covariate shift ($D = 0.73$, un-gated)	−0.003	+0.001	0.99	NO-STRONG-CONCEPT ✓—the null is load-bearing, not gated away
stationary (regression)	−0.012	−0.002	0.99	NO-STRONG-CONCEPT ✓
covariate shift, fixed *linear* rule (reg.)	+0.002	+0.004	0.91	UNIDENT ✓ (never CONCEPT)
covariate shift, fixed *nonlinear* rule (reg.)	−0.005	−0.003	0.92	UNIDENT ✓
label-noise decay, fixed rule (the F1 killer)	**+0.021 fires**	+0.004	3.54	**NOISE-DRIFT-CONFOUNDED** ✓
label noise *growing*, fixed rule	−0.023	−0.019	0.14	NO-STRONG-CONCEPT ✓
x-dependent noise, decaying scale	**+0.025 fires**	+0.005	3.76	**NOISE-DRIFT-CONFOUNDED** ✓

concept), earned blindness (learnable but unrecoverable ⇒ the geometry genuinely hides this signal class), or vacuity (no certificate). On the real audit, this distinction is load-bearing: 3 of 4 "unidentifiable" industrial cells fail the learnability gate.

3.3 Decision cascade and scope

The pre-registered cascade (PREREG §3; verbatim in the artifact): DEPLOYMENT-CONCEPT requires the *denoised* arm to fire (CI lower bound > 0 and mean > 0.02) within the noise envelope; a raw fire with the gate on is NOISE-DRIFT-CONFOUNDED; a raw fire with the gate off is RAW-ONLY-POSITIVE (unresolved, never concept); nulls route through D to the injection certificates; a sub-floor denoised CI is reported as a no-evidence band (calibrated: 2/5 no-drift anchor streams land there at 1/10–1/20 of the floor—CI-significant, decision-irrelevant). A strict secondary rule (CI lower bound > floor) is computed alongside; any cell whose verdict differs between the two readings is marked rule-sensitive and barred from headlines. Every verdict is scoped to the deployed hypothesis class: the ground-truth suite passes under HGB and random forests and fails under linear/kNN probes (§4), so tree-ensemble verdicts are decision-grade and linear/kNN run as canaries. All runs emit provenance metadata (commit, argv, library versions, seeds) into versioned artifacts; the decision rules, thresholds, seed protocol (exploratory 0–9, confirmatory 100–109), and aggregate reading were committed before the runs they govern.

4 Validation: the battery, the envelope, and the class matrix

4.1 The 14-cell pre-registered battery

The instrument must pass a synthetic ground-truth battery before any real-data run (PREREG §4); the battery is itself the executed record of the failure anatomy. All cells: $n = 12{,}000$, $d = 10$, $K = 10$, 5 seeds; verdicts under the full cascade (Table 2).

Three cells carry the paper's argument. The noise-decay cell is the executed refutation of "correct labels cannot hurt": the raw probe fires at a magnitude (+0.021) that matched a real industrial positive to within 0.003. The x-dependent-noise cell is a *second* raw false-positive channel, discovered during adversarial review of the repair itself; a third (noise scale correlated with the rule feature—the denoiser's worst case, since pseudo-label error then concentrates where the signal lives) was constructed by an independent red-team pass and also defused (raw +0.026 fires; denoised +0.006, null). The concept-plus-noise cell establishes that the repair keeps power where both mechanisms co-occur: denoised retention is 58% of the clean rotating-rule signal (+0.316 vs +0.541), and the same holds at small rule-change magnitudes (a 0.8-rad rotation: clean

Table 3: The model-class matrix: the five core controls with the probe's model class swapped.

control (truth)	HGB	RandomForest	MLP (64,32)	linear	kNN
rotating rule	✓ (+0.280)	✓ (+0.287)	✓ (+0.305)	✓ (+0.310)	✓ (+0.295)
covariate shift, fixed rule	✓ (−0.001)	✓ (+0.004)	△ (den +0.045, CI-saved)	✓ (−0.002)	× **CONCEPT (+0.098)**
prior shift, fixed rule	✓ (−0.003)	✓ (+0.003)	× **CONCEPT (den +0.042)**	× **CONCEPT (+0.026)**	× **CONCEPT (+0.048)**
stationary	✓	✓	✓	✓	✓
rotating rule + nuisance	✓ (+0.203)	✓ (+0.200)	✓ (+0.228)	✓ (+0.229)	✓ (+0.219)

denoised +0.108, with noise decay +0.062—attenuated, still firing). Robustness of the denoiser to a small old window was checked directly (old window capped at 600 rows, cross-fit folds of 300): no false positive (+0.004) and no false negative (+0.430).

Floor comparability across metrics. The 0.02 decision floor is shared across AUC, accuracy, and z-scored −RMSE, which are not decision-equivalent units. Two mitigations are structural: every null carries its own detectable-effect bound δ (the per-dataset, per-metric power statement that does the aggregate's real work), and CONCEPT never rests on the floor alone (the denoised CI, gate, and envelope must all align). As a sensitivity check, re-thresholding all committed runs under per-metric floors rescaled by the battery's matched-strength concept magnitudes (binary +0.28 vs regression +0.55 $\Rightarrow$ regression floor 0.04) moves no cell across the concept/no-concept boundary; the only affected cell is the mechanism label of the diagnosed regression positive (§5.3), which is already flagged rule-sensitive and barred from headlines.

4.2 The envelope: where the repair itself breaks

The denoiser is biased—pseudo-label error grows with old-window noise—and rather than assuming the bias negligible we mapped it: on pure fixed-rule nulls, the denoised arm reads +0.0037 at noise-ratio 3.54, +0.0048 at 3.76, +0.0043 at 3.87, +0.0140 at 4.72, and +0.0259 **at 5.71—across the 0.02 decision floor on a null**. The instrument therefore abstains (NOISE-AMBIGUOUS) above ratio 4.7. Separation still exists beyond the envelope (at ratio ≈ 6, a true rotating rule reads denoised +0.086 vs the null's +0.026, disjoint CIs), but any threshold there would be calibrated on a single noise family, so we refuse the verdict instead. Stable controls calibrate the gate statistic at 0.75–0.99, far from the 1.5 firing threshold; every real-data gate value in §5 falls either clearly below 1.5 or in 2.0–2.9—nowhere near the envelope edge.

4.3 The model-class matrix: why verdicts are class-scoped

Re-running the five core controls with the probe's model class swapped (the instrument's every component—staleness arms, gate, separability, injection—follows the swap) yields Table 3.

Concept *detection* is class-robust (all five classes fire on true rotations); the concept/covariate *separation*—the property the verdict rests on—holds only for classes flexible enough to represent the fixed rule. This is Shimodaira (2000) made empirical: under misspecification the mixture-ERM optimum moves with $P(x)$, so covariate shift alone manufactures "old data hurts." Denoising does not repair this channel (kNN still false-fires at +0.111 with pseudo-labels—the misspecification is in the *probe*, not the labels), which is why the instrument's verdicts are stated as tree-ensemble-scoped, with linear/kNN carried as canaries. The neural probe deserves emphasis, because the paper's motivating question is whether deep tabular architectures have a rule change to exploit: **a two-layer MLP probe fails the separation battery**—it reads fixed-rule prior shift as concept drift (denoised +0.042, a fire) and leans positive on pure covariate shift (denoised +0.045, saved only by CI width)—so it joins the canaries rather than the decision-grade classes. We do not claim this extrapolates to modern deep tabular architectures at scale; we claim the direction of the burden: a probe class earns drift-verdict authority by passing this battery, and the first neural probe we tested does not. The practical reading for the field is uncomfortable: **a drift monitor built on a linear, local, or small-neural probe can report concept drift on data where a tree-ensemble monitor reports**

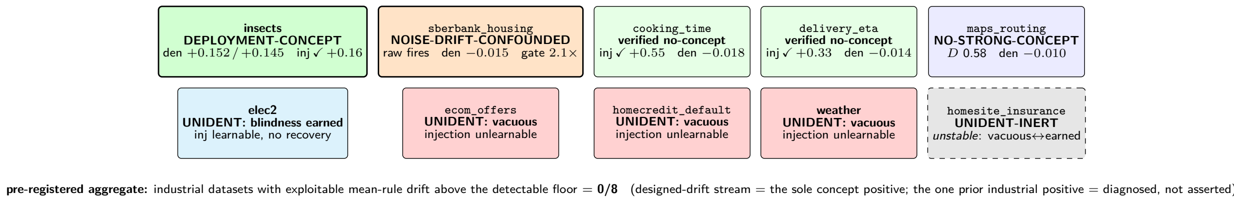

Figure 2: The identifiability map at a glance (Table 4 is the precise record). Tile color encodes the verdict class: green = rule-change verdict or verified no-concept; amber = diagnosed noise mechanism; blue = identifiable null; light blue = earned blindness; red = certificate refused; grey (dashed) = unstable across seed sets. Each tile carries its load-bearing number (denoised staleness, exploratory / confirmatory where both exist, or injection recovery).

Table 4: The identifiability map. Where two numbers are shown: exploratory / confirmatory.

dataset	verdict (= confirmatory)	raw	denoised	gate	certificate
insects	**DEPLOYMENT-CONCEPT**	+0.135 / +0.129	**+0.152 / +0.145**	1.24	injection recovers (+0.162)
`sberbank_housing`	**NOISE-DRIFT-CONFOUNDED**	+0.024 / +0.033 (fires)	**−0.015 / −0.011**	**2.11 / 2.21 (fires)**	— (diagnosed, §5.3)
`cooking_time`	INJECTION-RECOVERED	−0.011	−0.018	0.92	**verified no-concept** (inj +0.55)
`delivery_eta`	INJECTION-RECOVERED	−0.012	−0.014	0.89	**verified no-concept** (inj +0.33)
`maps_routing`	NO-STRONG-CONCEPT	−0.008	−0.010	1.01	identifiable region (D 0.58)
elec2	UNIDENTIFIABLE	+0.001	+0.001	0.66	**blindness earned** (inj learnable, +0.018)
`ecom_offers`	UNIDENTIFIABLE	−0.001	−0.007	1.25	*vacuous*—injection unlearnable
`homecredit_default`	UNIDENTIFIABLE	−0.007	+0.004	1.29	*vacuous* (80 proxy features stripped)
weather	UNIDENTIFIABLE	−0.012	−0.011	0.93	*vacuous*
`homesite_insurance`	UNIDENTIFIABLE-INERT	−0.004	−0.002	1.14	*unstable* (vacuous↔earned across seed sets)

none, on the same bytes—and §5.4 shows this happens on the canonical real drift dataset, not just in synthetics.

5 The Map

5.1 Protocol

Eight TabReD datasets (Rubachev et al., 2025, train segments of the official temporal splits), Electricity (elec2), and INSECTS (incremental-balanced)—HGB probe, $K = 10$, 10 exploratory seeds (0–9), then a **confirmatory rerun with fresh seeds (100–109)**; any verdict that moves between the two is reported unstable and barred from claims. Decision cascade, thresholds, seed protocol, and the aggregate reading were committed before execution; the prior (v2-era) industrial positive was pre-registered as a *retraction candidate with a prediction* (NOISE-DRIFT-CONFOUNDED or denoised-null), and a survival battery was pre-specified in case the prediction failed. Both the primary rule (CI $> 0 \wedge$ mean $>$ floor) and the strict rule (CI $>$ floor) are computed; rule-sensitive cells are flagged.

5.2 Results: 10/10 confirmatory-stable; industrial mean-rule drift 0/8

Figure 2 gives the map at a glance; Table 4 is the precise record. The pre-registered aggregate reading: **relative to the tree-ensemble class, the number of industrial datasets with exploitable mean-rule drift above the per-dataset detectable floor is 0/8.** The only concept positive is the designed-drift stream, where the denoised arm *exceeds* the raw arm—pseudo-labels encode the old rule cleanly once label noise is removed, the signature of genuine rule change. Three of four blindness claims that a naive protocol would have granted turn out to be *vacuous* (the planted probe rule is unlearnable in those geometries—their top-variance features cannot carry it), a distinction invisible without the learnability gate. The strongest cells in the map are, counter-intuitively, the two verified no-concept certificates: geometry demonstrably has power (+0.55/+0.33 recovery of a planted rule) and the real staleness is still null.

5.3 The diagnosis: how the instrument caught its own prior positive

The v2-era instrument (raw arm only) had reported sberbank-housing—the panel's one regression dataset—as its sole industrial concept positive (+0.024 $[+0.018, +0.030]$). The audit constructed the noise-decay null of §4.1 at matching magnitude; the v3 rerun then delivered the diagnosis. Across window resolutions $K \in \{5, 8, 10, 12, 20\}$: the raw arm fires at $K \in \{8, 10, 12\}$ (at $K = 10$ reproducing the v2 headline **bit-for-bit**[1] at +0.0239—the raw pipeline's RNG stream is unchanged, so this is the *same* signal reinterpreted, not a re-measurement that happened to differ); the old window's measured noise proxy is 2.1–2.9× the recent median at every K; and the **denoised arm is significantly negative at every** K (-0.014 to -0.018, all CIs below zero): replace the old labels with cross-fitted pseudo-labels and the old rows *help*. The rule did not change; the early labels are noisier—consistent with 2011–2012 crisis-era Russian housing prices. At $K = 20$ the injection control recovers a planted rule (+0.101) through the same geometry, and at no K does any rule reading yield CONCEPT. We are not aware of a prior case of a drift-attribution instrument diagnosing the *mechanism* of its own earlier false positive on real data, as opposed to merely failing to replicate it.

5.4 The class panel on real data

The decision-grade classes agree: **HGB and random-forest verdicts match operatively on 10/10 datasets** (one sub-label difference). The canaries do not—and the flip that matters is replicated across two independent canary classes: **both the linear probe (raw +0.023, denoised +0.033, injection recovers +0.190) and the MLP probe (raw +0.007, denoised +0.025) read Electricity as DEPLOYMENT-CONCEPT** where the tree-ensemble probe reads it unidentifiable-with-earned-blindness. Electricity is the literature's canonical concept-drift dataset; whether a monitor confirms that reputation depends on the probe class, on identical data. The MLP panel also illustrates why canary verdicts must not be consumed directly: on the regression datasets its staleness estimates are numerically unstable (sberbank CI spanning ±15 z-units; weather CI width 0.5)—driven by a few divergent fits: on sberbank, three of ten seeds land at $|\text{staleness}| > 10$ z-units while the other seven lie within ±2 (per-seed values in the committed artifact), i.e., occasional optimization failure of the un-tuned probe rather than signal—and no TabReD cell produces a new positive under it—the decision-grade map is unchanged. A second canary behavior is worth reporting as a warning: on `ecom_offers` the linear probe fires the *denoised* arm only (+0.028) with the raw arm null (-0.002)—a denoiser-artifact channel specific to misspecified probes (the linear $\hat{g}_{\text{old}}$'s systematic error is itself informative to the linear downstream model), reinforcing that the denoised arm's semantics are also class-scoped.

6 Anchors and the Sensitivity Profile

The map's credibility rests on knowing what the instrument *can* see. We ran it over 23 synthetic river streams (SEA/Agrawal/STAGGER/Sine/Hyperplane; no-drift / abrupt single-switch / gradual / reoccurring variants) and all seven INSECTS variants (real sensor data, lab-controlled temperature drift; Souza et al., 2020).

Monotone and single-switch rule changes fire: 9/9. River: `agrawal_abrupt` +0.045, `agrawal_gradual` +0.047, `stagger_abrupt`, `sine_abrupt` +0.031, `hyperplane_incremental` +0.113 (with the gate correctly co-flagging its noise component), `sine_reoccur2`[2] +0.047. INSECTS: gradual-balanced

[1] Bit-identical at full precision: +0.023900573591226625, recorded verbatim in the committed artifact.

[2] Reoccurring in name only for this lens's horizon: the stream returns to its initial regime just in its final ≈22%, so most evaluated back-half windows sit inside the middle regime, whose rule is a near label-inversion of the anchor's—its recency gain

+0.092, gradual-imbalanced +0.069, incremental +0.135—in every firing cell, denoised $\geq$ raw, and the injection positive-control recovers. Weak switches (SEA's threshold nudge) land in the no-evidence band with consistent sign, reported with their detectable floors.

Recurring regimes are correctly silent—and fingerprinted. INSECTS abrupt variants (oscillating temperature) read *negative* staleness (old data helps: −0.070, −0.027) with **negative recency gain** (−0.058, −0.023): the window adjacent to the test predicts it *worse* than the oldest window does. Negative recency is impossible under one-way drift; it is the signature of a regime that has returned. The same pattern holds across river's reoccurring cells and INSECTS incremental-reoccurring—three independent confirmations. This is a scope statement, not a defect: the lens answers the deployment question ("does old data harm a model trained today?"), and when old regimes recur, old data genuinely does not harm—recency does. A monitor built on this lens will not *detect* recurring concept drift; it will correctly tell you old data is safe to keep, and its negative-recency flag tells you why.

Calibration of the no-evidence band. Two of five no-drift anchor streams produce "CI-significant" sub-floor denoised positives at 1/10–1/20 of the decision floor—seed-level CIs on overlapping subsamples are anti-conservative at tiny magnitudes. Accordingly, the sub-floor band is read as *no evidence* everywhere in this paper, a calibration the anchor suite forced and the pre-registration records.

Malware (EMBER). On 2018-dense monthly windows the cell is certificate-grade: DEPLOYMENT-DECAY-COVARIATE with $D = 0.834$ (identifiable—the null carries full weight), raw staleness −0.008 [−.009, −.007] (old data helps), denoised −0.004, gate quiet, recency gain +0.031 above the floor, detectable floor 0.0013. Malware's temporal degradation (Pendlebury et al., 2019) is real and recency-recoverable, but it is coverage-driven—new families appear; old labels do not rot—and the deployment lens draws exactly that distinction: keep the old data, expect decay anyway, retrain for coverage. (A full-history windowing over 126 sparse months gives the same null at lower power, and a mis-windowed attempt on 19-row windows correctly refused to emit a verdict.)

The WhyShift bridge (ACS across years). WhyShift reported $Y \mid X$-shifts as prevalent on ACSIncome across *states*; running the instrument on the same task across *years* (California, 2014–2018, yearly windows, 10 seeds) yields the map's best-powered null: NO-STRONG-CONCEPT with raw −0.008 and denoised −0.007 (both CIs negative—old years help), gate quiet, $D = 0.515$ (the years are barely separable in feature space—negligible covariate movement, fully identifiable), recency ≈ 0, and a detectable floor of 0.0008. The same task whose spatial axis exhibits prevalent $Y \mid X$-shift is temporally quiet over five years—the axis, not just the dataset, determines the shift type. (An overlap-based decomposition run on the same cell agrees: within-overlap gap −0.009 $\approx$ placebo −0.008 at cov-AUC 0.68—Appendix B.) The cell also passes a real-data analogue of the prior-shift control: the fixed \$50k income threshold under inflation produces a monotone positive-rate ramp (0.36 $\rightarrow$ 0.42), and the instrument does not misread that prior drift as concept. (Scope: one state, five pre-COVID years; a multi-state extension is future work.)

7 Discussion

What a practitioner gets. Raw staleness answers "does old data hurt?"; the repaired instrument answers "*why*, and what to do about it," with a decision rule per cell: a denoised fire inside the envelope $\Rightarrow$ the rule changed for your model class—old labels are poison, retrain recent-only or add temporal structure; a raw fire with the gate on $\Rightarrow$ your labels' noise profile drifted—old rows are *reusable* via self-training (the denoised arm is literally that remedy, measured); nulls with a recovery certificate $\Rightarrow$ retraining chases coverage, not rule change; negative recency $\Rightarrow$ regimes recur—retention, not recency, is your friend; vacuous or unstable certificates $\Rightarrow$ do not let a monitor speak about this dataset's drift type at all.

What the map means, and does not. The 0/8 aggregate does *not* say concept drift is absent from industrial tabular ML: three cells are certificate-less and one is blind-but-certified; blindness is not absence. It

is strongly *positive* (+0.30), the opposite of the recurring fingerprint, and geometry-matched siblings whose middle regime is not label-inverting read $\approx$0 (`sine_reoccur` −0.003, `sine_reoccur3` −0.003).

says something narrower and, we argue, more useful: on the benchmark the field uses to argue for time-aware tabular architectures, **no measurable mean-rule drift exists for the model class that dominates those benchmarks, and every apparent counterexample so far has dissolved under controls**—the last one into label-noise decay, diagnosed rather than asserted. The burden of proof for "temporal architecture X exploits concept drift on TabReD-like data" should now include an identifiability certificate.

Limitations. (1) Verdicts are tree-ensemble-scoped; we show this scoping is necessary, not that it is sufficient for every deployed system. The panel's neural probe (a two-layer MLP) fails the separation battery and is classified a canary (§4.3); modern deep tabular architectures (FT-Transformer-class, at production scale) remain untested—extending the panel is the clearest next step, with the battery as the pre-registered entry bar any such probe must pass before its drift verdicts are trusted. (1b) **Scale.** The probe trains on $N \leq 6{,}000$ rows per arm; industrial models train on orders of magnitude more. A rule change exploitable only at much larger N is invisible here, so every δ bound and the 0/8 aggregate are statements *at probe scale*. A first $\delta(N)$ sweep (Appendix B) raises the arm cap to the window-geometry ceiling ($N \approx$ 14k–24k, up to 4× the headline scale) on the three largest null cells: every verdict is unchanged and no reading trends toward the floor. Production-scale N beyond that remains future work, and we flag rather than dismiss the possibility that the map changes there. (2) $K = 10$ rolling windows with a fixed early anchor: drift at time scales far below the window width is averaged away (the injection control partially measures this—its recovery varies with K), and the TabReD map covers the train segments of the official splits, not the held-out deployment gap. (3) The single robust positive is lab-designed drift; we found no naturally occurring industrial positive to calibrate magnitude against—that is the map's finding and its weakness simultaneously. (4) The envelope constant (4.7) is calibrated on Gaussian noise families; heavy-tailed label noise inherits only the abstention discipline, not the exact boundary. (5) TabReD is curated to be temporally splittable; external validity to industrial data at large is an inductive step we flag rather than take.

On process. This project's ledger records nine positive findings that dissolved under scrutiny before the present result; the ninth dissolution is §5.3, and it is the only one the measuring instrument itself diagnosed. The methodological claim we stand behind is that the combination that finally produced a stable result—executed adversarial nulls before real-data claims, pre-registered cascades with commit-timestamped predictions, confirmatory fresh-seed replication, certificates instead of assumptions, and canary probes—is cheap relative to the cost of the dissolutions it prevents, and we release the audit trail as part of the artifact.

Future work. Multi-state and post-2019 extensions of the ACS bridge (§6); class-invariance conditions for the denoised arm (when does a probe family admit *any* sound staleness reading?); a $\delta(N)$ scaling study; and porting the certificate protocol to streaming monitors, where the recurring-regime fingerprint (negative recency) is directly actionable.

8 Reproducibility

Everything is in one repository, linked in anonymized form for review. The instrument is a single sklearn-only script (`scripts/run_deployment_decay.py`); every stochastic step is seeded, and every output artifact embeds its commit hash, argv, library versions, and UTC timestamp.

Compute. Every run in the paper is CPU-only—the instrument and all probe classes are scikit-learn models; no GPU is used anywhere—executed single-node on one shared multicore Linux server (Python 3.11.15, scikit-learn 1.9.0, NumPy 2.4.6; the environment freeze is committed). Wall-clock is reconstructable from the committed phase logs: the main pre-registered phases ran phase-parallel in ≈17 h on one calendar day, the model-class panel adds ≈2 h, and the optional cells (EMBER, the ACS bridge, canary reruns) ≈13 h across two further days; no single dataset cell exceeds a few CPU-hours. Per-source data access and license terms are tabulated in Appendix A. The synthetic battery is byte-reproducible (re-running `--synth` regenerates the committed artifact SHA-identical on the same environment), and the raw arm's RNG stream is stable across instrument versions (the v2 headline number reproduces bit-for-bit under v3, which is what makes §5.3 a reinterpretation rather than a re-roll). The pre-registration (`PREREG_DEPLOYMENT_V2.md`) freezes thresholds, cascade, seed protocol, and aggregate reading, with results appended as read-only sections whose ordering is enforced by the git history; all server runs are reproduced by one resumable driver (`scripts/run_prereg_phases.sh`, marker-based, phase-parallel), and the environment freeze of the machine that produced the headline numbers is committed. Data access: TabReD requires Kaggle authentication and per-competition rule acceptance; elec2 fetches from OpenML; INSECTS and the river panel install via `river`; EMBER-2018 downloads from its public archive and is parsed by a dependency-free adapter. Exact cross-version bit-reproducibility of HistGradientBoosting across sklearn releases is *not* claimed; verdict-level stability across seed sets and across HGB/RF is (10/10 and 10/10, §5).

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A Data access and licenses

Verified against each distributor at the time of writing (Table 5).

Table 5: Data access paths and license terms, per source.

source	access path in the pipeline	license / terms
TabReD (8 industrial datasets)	Kaggle, via the TabReD preprocessing tooling	per-competition Kaggle rules (acceptance required); TabReD tooling Apache-2.0
Electricity (elec2)	OpenML dataset 151	listed "Public" by OpenML
INSECTS (7 variants)	fetched through the `river` package	`river` BSD-3-Clause; dataset introduced by Souza et al. (2020)
river synthetic streams (SEA / Agrawal / STAGGER / Sine / Hyperplane)	generated by `river`	BSD-3-Clause
EMBER-2018	public archive download; parsed by our dependency-free adapter	data files MIT (the ember *code* is AGPL-v3 and is not used)
ACS (folktables bridge)	`folktables`	folktables MIT; ACS PUMS is public U.S. Census Bureau data

B N-scaling and cross-lens agreement (post-hoc robustness checks)

Both checks were run after the pre-registered audit froze, with the instrument, cascade, and thresholds unchanged (commit-stamped artifacts); they are robustness checks, not pre-registered cells.

B.1 $\delta(N)$: the map does not move toward the floor as N grows. The probe's headline scale is $N \leq 6{,}000$ (§7, Limitation 1b). We re-ran the three largest null cells with the arm cap swept over $\{1{,}500,\ 24{,}000,\ 96{,}000\}$ (10 seeds each). Window geometry bounds the realizable arm size ($N = \min(|\text{recent}|, |\text{old}|, \text{cap})$, and $|\text{window}| \approx n/K$), so the realized ceilings are ≈17,200 on `homecredit_default` (the 96,000 cap realizes the same N), 24,000 on weather (cap-bound), and ≈14,300 on `maps_routing`. Verdicts are unchanged at every N, and nothing trends toward the 0.02 floor (Table 6). Raw staleness is significantly *negative* (old data helps) at every N on all three cells; the one sub-floor positive (homecredit's denoised arm) sits at a quarter of the decision floor, is flat across an 11× range of N, and shrinks at the top of the range; weather's denoised arm converges to ≈0 from below. The injection certificates reproduce (homecredit/weather planted rules remain unlearnable—vacuous; maps identifiable). Scope unchanged: $K = 10$ window geometry caps realizable N at ≈14k–24k here, so production-scale N beyond that remains untested.

Table 6: $\delta(N)$ sweep on the three largest null cells (10 seeds; rows at $N = 6{,}000$ are the main-audit values of Table 4).

cell	realized N	raw staleness	denoised	verdict
`homecredit_default`	1,500	-0.015 $[-.018, -.012]$	$+0.007$ $[+.005, +.008]$	UNIDENT (vacuous)—unchanged
`homecredit_default`	6,000	-0.007	$+0.004$	unchanged
`homecredit_default`	≈17,200 (cap 24k)	-0.013 $[-.016, -.011]$	$+0.006$ $[+.004, +.007]$	unchanged
`homecredit_default`	≈17,200 (cap 96k)	-0.010 $[-.012, -.009]$	$+0.005$ $[+.003, +.006]$	unchanged
weather	1,500	-0.017 $[-.019, -.015]$	-0.022 $[-.024, -.020]$	unchanged
weather	6,000	-0.012	-0.011	unchanged
weather	24,000	-0.005 $[-.006, -.005]$	-0.002 $[-.002, -.002]$	unchanged
`maps_routing`	1,500	-0.009 $[-.010, -.009]$	-0.017 $[-.017, -.016]$	NO-STRONG-CONCEPT
`maps_routing`	6,000	-0.008	-0.010	unchanged
`maps_routing`	≈14,300 (cap 24k)	-0.008 $[-.009, -.008]$	-0.008 $[-.009, -.008]$	unchanged

B.2 Cross-lens agreement where overlap holds. On the two cells of the map whose separability leaves an overlap-based decomposition applicable (`maps_routing`, $D = 0.58$; the ACS bridge, $D = 0.515$), we ran the WhyShift-style within-overlap concept decomposition (ess-gated) next to the deployment lens. The two instruments agree (Table 7): where overlap holds, the overlap lens and the deployment lens return the same verdict; the instrument disagreements this paper documents (§3–4) arise precisely where overlap fails—which is why verdicts must carry certificates.

Table 7: Cross-lens agreement on the overlap-valid cells.

cell	deployment lens	within-overlap gap [CI]	note
`maps_routing` (full, 986 feats)	NO-STRONG-CONCEPT (den -0.010)	-0.003 $[-.004, -.002]$	measurable 5/5 seeds, ESS 93%
`maps_routing` (sparse MI@5/10/20/50)	unchanged	$-0.005 \ldots -0.005$, all ≈ 0	measurable under every representation
ACS CA 2014→2018	NO-STRONG-CONCEPT (den -0.007)	-0.009 (placebo -0.008)	cov-AUC 0.68; gap $\approx$ placebo